# Model Architecture: Qwen2.5-0.5B-Instruct

Generated by sg_modelload

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# 1 Model Overview

| | |
|---|---|
| Model name | Qwen2.5-0.5B-Instruct |
| Transformers class | `Qwen2ForCausalLM` |
| Path | `/models/sayak_da_work/Qwen2.5-0.5B-Instruct` |
| Total parameters | 494,032,768 (494.03M) |
| Weight dtype | bfloat16 |
| Hidden size $d$ | 896 |
| Decoder layers $L$ | 24 |
| Query heads $H$ | 14 |
| Key/value heads $H_{kv}$ | 2 |
| MLP width $d_{ff}$ | 4,864 |
| Vocabulary $V$ | 151,936 |
| Max context length | 32,768 |
| RMSNorm $\epsilon$ | 1e-06 |
| Tied input/output embeddings | True |
| Activation | SiLU |
| RoPE base $\theta$ | 1,000,000 |

**How to read this document.** Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

### Notation

| | |
|---|---|
| $B$, $T$ | batch size, sequence length (tokens) |
| $d$ | hidden size (896) |
| $V$ | vocabulary size (151,936) |
| $H$, $H_{kv}$, $d_h$ | query heads, key/value heads, size of one head |
| $W_*$, $b_*$ | learned weight matrix / bias of the named projection |
| $\odot$, $\sigma$ | element-wise product, logistic sigmoid |
| $f^{\circ n}$ | $f$ applied $n$ times in a row ($f \circ f \circ \cdots \circ f$) |
| $g \circ f$ | first $f$, then $g$ (composition reads right to left) |
| RMSNorm$(x)$ | $\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale $\gamma$) |
| LN$(x)$ | LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$ |
| RoPE | rotary position embedding (rotates query/key pairs by an angle set by position) |

# 2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 24 decoder layers are written with composition powers, so a block that repeats $n$ times appears once as $\mathcal{B}^{\circ n}$.

$$
\begin{aligned}
h_0 &= \text{Embed}(x) \\
h_L &= \left(\mathcal{B}_{\text{F}}^{\circ 24}\right)(h_0) \\
y &= W_E^{\top}\ \text{RMSNorm}_{\text{final}}(h_L)
\end{aligned}
$$

## 2.1 Layer pattern

| Symbol | Block type | Count | Params / block | Layer indices |
|---|---|---|---|---|
| $\mathcal{B}_{\text{F}}$ | Full (global) attention block | 24 | 14,912,384 | 0–23 |

**Language-model decoder (24 layers).** layers 0–23: $[\mathcal{B}_{\text{F}}]$ repeated 24×.

## 2.2 Block functions

$\mathcal{B}_{\text{F}}$ — **Full (global) attention block.**

$$a = h + \text{Attn}(\text{RMSNorm}_1(h)) \qquad \text{pre-norm + residual}$$

$$
\begin{aligned}
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
Q &= W_Q\,u + b_Q && H = 14 \text{ heads of size } d_h = 64\\
K,\ V &= W_K\,u + b_K, \quad W_V\,u + b_V && H_{kv} = 2 \text{ key/value heads}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\,Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H]\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\mathrm{SiLU}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 4864}
\end{aligned}
$$

# 3 Forward Pass in Pseudocode

**Algorithm 1** Qwen2.5-0.5B-Instruct: full forward pass

**Require:** token ids $x$ of shape $B \times T$

$h \leftarrow \mathrm{Embed}(x)$ ▷ look up one vector per token: $B{\times}T{\times}896$
**for** $r \leftarrow 0$ **to** 23 **do** ▷ layers 0–23: pattern of 1 repeated 24 times
  $h \leftarrow \mathcal{B}_{\mathrm{F}}(h)$ ▷ Full (global) attention block
**end for**
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$
$y \leftarrow h\,W_E^{\top}$ ▷ logits $B{\times}T{\times}151,936$, embedding matrix re-used
**return** $y$ ▷ softmax$(y)$ = next-token probabilities

**Algorithm 2** Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

**Require:** $h$ of shape $B \times T \times 896$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
*token mixing: multi-head self-attention*
$Q \leftarrow uW_Q^{\top} + b_Q$ ▷ $B{\times}T{\times}14{\times}64$ (14 query heads)
$K \leftarrow uW_K^{\top} + b_K;\ V \leftarrow uW_V^{\top} + b_V$ ▷ $B{\times}T{\times}2{\times}64$ (2 key/value heads, each shared by 7 query heads)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^{\top} + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}14{\times}64$
$m \leftarrow O\,W_O^{\top}$ ▷ concatenate heads, project back to $B{\times}T{\times}896$
$h \leftarrow r + m$ ▷ add residual
*channel mixing: feed-forward network (each token independently)*
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^{\top}) \odot uW_{\mathrm{up}}^{\top}\big)W_{\mathrm{down}}^{\top}$ ▷ expand to 4864, gate, shrink back to 896
$h \leftarrow r + m$ ▷ add residual
**return** $h$

# 4 Compact Architecture (repeated layers written once)

## 4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

| Weight | Type | Shape | Params |
|---|---|---|---|
| `model.embed_tokens.weight` | Embedding | 151936 × 896 | 136,134,656 |
| `model.norm.weight` | Qwen2RMSNorm | 896 | 896 |
| `lm_head.weight` (tied to `model.embed_tokens.weight`) | Linear | 151936 × 896 | 0 (shared) |

## 4.2 Language-model decoder: `model.layers`

Table 3: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 0–23; weight names are relative to `model.layers.`$\ell$

| Weight (relative) | Type | Shape | Params |
|---|---|---|---|
| `self_attn.q_proj.weight` | Linear | $896 \times 896$ | 802,816 |
| `self_attn.q_proj.bias` | Linear | 896 | 896 |
| `self_attn.k_proj.weight` | Linear | $128 \times 896$ | 114,688 |
| `self_attn.k_proj.bias` | Linear | 128 | 128 |
| `self_attn.v_proj.weight` | Linear | $128 \times 896$ | 114,688 |
| `self_attn.v_proj.bias` | Linear | 128 | 128 |
| `self_attn.o_proj.weight` | Linear | $896 \times 896$ | 802,816 |
| `mlp.gate_proj.weight` | Linear | $4864 \times 896$ | 4,358,144 |
| `mlp.up_proj.weight` | Linear | $4864 \times 896$ | 4,358,144 |
| `mlp.down_proj.weight` | Linear | $896 \times 4864$ | 4,358,144 |
| `input_layernorm.weight` | Qwen2RMSNorm | 896 | 896 |
| `post_attention_layernorm.weight` | Qwen2RMSNorm | 896 | 896 |
| **Per block** | | | **14,912,384** |
| **$\times$ 24 blocks** | | | **357,897,216** |

# 5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

| Layer (appears as) | What it does |
|---|---|
| LANGUAGE MODEL | |
| **Token embedding**<br>`embed_tokens` | Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$). |
| **Query projection (q_proj)**<br>`self_attn.q_proj` | Turns each token vector into query vectors, one per head: "what am I looking for?". |
| **Key projection (k_proj)**<br>`self_attn.k_proj` | Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads. |
| **Value projection (v_proj)**<br>`self_attn.v_proj` | Turns each token into value vectors: the information actually copied to tokens that attend to it. |
| **Output projection (o_proj)**<br>`self_attn.o_proj` | Concatenates all heads' results and mixes them back into one hidden-size vector per token. |
| **Gate projection (gate_proj)**<br>`mlp.gate_proj` | Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit. |
| **Up projection (up_proj)**<br>`mlp.up_proj` | Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU). |
| **Down projection (down_proj)**<br>`mlp.down_proj` | Shrinks the gated hidden vector back to the model width so it can be added to the residual stream. |
| **Input norm (input_layernorm)**<br>`layers.input_layernorm` | RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight. |
| **post_attention_layernorm**<br>`layers.post_attention_layernorm` | Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP. |
| **Final norm**<br>`norm` | RMSNorm applied once after the last decoder layer, just before the LM head. |

# 6 Parameter Count

Table 5: Parameters by component

| Component | Parameters | % of total |
|---|---|---|
| MLP / MoE | 313,786,368 | 63.5% |
| Token embedding | 136,134,656 | 27.6% |
| Attention / token mixer | 44,067,840 | 8.9% |
| Norms | 43,904 | 0.0% |
| **Total** | **494,032,768** | **100.0%** |

Table 6: Parameters by block type

| Block type | Count | Params / block | Total |
|---|---|---|---|
| $\mathcal{B}_{\mathrm{F}}$ Full (global) attention block | 24 | 14,912,384 | 357,897,216 |
| Modules outside stacks | 1 | – | 136,135,552 |
| **Total** | | | **494,032,768** |

# Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

| Layer path | Type | Shape | Dtype | Params |
|---|---|---|---|---|
| `lm_head.weight` *(tied)* | Linear | 151936 × 896 | bfloat16 | 136,134,656 |
| `model` | Qwen2Model | – | – | – |
| `model.embed_tokens.weight` | Embedding | 151936 × 896 | bfloat16 | 136,134,656 |
| `model.layers.0` | Qwen2DecoderLayer | – | – | – |
| `model.layers.0.input_layernorm.weight` | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| `model.layers.0.mlp` | Qwen2MLP | – | – | – |
| `model.layers.0.mlp.down_proj.weight` | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| `model.layers.0.mlp.gate_proj.weight` | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| `model.layers.0.mlp.up_proj.weight` | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| `model.layers.0.post_attention_layernorm.weight` | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| `model.layers.0.self_attn` | Qwen2Attention | – | – | – |
| `model.layers.0.self_attn.k_proj.bias` | Linear | 128 | bfloat16 | 128 |
| `model.layers.0.self_attn.k_proj.weight` | Linear | 128 × 896 | bfloat16 | 114,688 |
| `model.layers.0.self_attn.o_proj.weight` | Linear | 896 × 896 | bfloat16 | 802,816 |
| `model.layers.0.self_attn.q_proj.bias` | Linear | 896 | bfloat16 | 896 |
| `model.layers.0.self_attn.q_proj.weight` | Linear | 896 × 896 | bfloat16 | 802,816 |
| `model.layers.0.self_attn.v_proj.bias` | Linear | 128 | bfloat16 | 128 |
| `model.layers.0.self_attn.v_proj.weight` | Linear | 128 × 896 | bfloat16 | 114,688 |
| `model.layers.1` | Qwen2DecoderLayer | – | – | – |
| `model.layers.1.input_layernorm.weight` | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| `model.layers.1.mlp` | Qwen2MLP | – | – | – |
| `model.layers.1.mlp.down_proj.weight` | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| `model.layers.1.mlp.gate_proj.weight` | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| `model.layers.1.mlp.up_proj.weight` | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| `model.layers.1.post_attention_layernorm.weight` | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| `model.layers.1.self_attn` | Qwen2Attention | – | – | – |
| `model.layers.1.self_attn.k_proj.bias` | Linear | 128 | bfloat16 | 128 |
| `model.layers.1.self_attn.k_proj.weight` | Linear | 128 × 896 | bfloat16 | 114,688 |
| `model.layers.1.self_attn.o_proj.weight` | Linear | 896 × 896 | bfloat16 | 802,816 |
| `model.layers.1.self_attn.q_proj.bias` | Linear | 896 | bfloat16 | 896 |
| `model.layers.1.self_attn.q_proj.weight` | Linear | 896 × 896 | bfloat16 | 802,816 |
| `model.layers.1.self_attn.v_proj.bias` | Linear | 128 | bfloat16 | 128 |
| `model.layers.1.self_attn.v_proj.weight` | Linear | 128 × 896 | bfloat16 | 114,688 |
| `model.layers.2` | Qwen2DecoderLayer | – | – | – |
| `model.layers.2.input_layernorm.weight` | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| `model.layers.2.mlp` | Qwen2MLP | – | – | – |
| `model.layers.2.mlp.down_proj.weight` | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| `model.layers.2.mlp.gate_proj.weight` | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| `model.layers.2.mlp.up_proj.weight` | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| `model.layers.2.post_attention_layernorm.weight` | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| `model.layers.2.self_attn` | Qwen2Attention | – | – | – |
| `model.layers.2.self_attn.k_proj.bias` | Linear | 128 | bfloat16 | 128 |
| `model.layers.2.self_attn.k_proj.weight` | Linear | 128 × 896 | bfloat16 | 114,688 |

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| Layer path | Type | Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.2.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.2.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.2.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.2.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.2.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.3 | Qwen2DecoderLayer | – | – | – |
| model.layers.3.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.3.mlp | Qwen2MLP | – | – | – |
| model.layers.3.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.3.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.3.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.3.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.3.self_attn | Qwen2Attention | – | – | – |
| model.layers.3.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.3.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.3.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.3.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.3.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.3.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.3.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.4 | Qwen2DecoderLayer | – | – | – |
| model.layers.4.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.4.mlp | Qwen2MLP | – | – | – |
| model.layers.4.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.4.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.4.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.4.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.4.self_attn | Qwen2Attention | – | – | – |
| model.layers.4.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.4.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.4.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.4.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.4.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.4.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.4.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.5 | Qwen2DecoderLayer | – | – | – |
| model.layers.5.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.5.mlp | Qwen2MLP | – | – | – |
| model.layers.5.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.5.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.5.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.5.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.5.self_attn | Qwen2Attention | – | – | – |
| model.layers.5.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.5.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.5.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.5.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.5.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.5.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.5.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.6 | Qwen2DecoderLayer | – | – | – |
| model.layers.6.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.6.mlp | Qwen2MLP | – | – | – |
| model.layers.6.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.6.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.6.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.6.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.6.self_attn | Qwen2Attention | – | – | – |
| model.layers.6.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.6.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.6.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.6.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.6.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.6.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.6.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.7 | Qwen2DecoderLayer | – | – | – |
| model.layers.7.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.7.mlp | Qwen2MLP | – | – | – |
| model.layers.7.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.7.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |

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| Layer path | Type | Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.7.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.7.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.7.self_attn | Qwen2Attention | – | – | – |
| model.layers.7.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.7.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.7.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.7.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.7.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.7.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.7.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.8 | Qwen2DecoderLayer | – | – | – |
| model.layers.8.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.8.mlp | Qwen2MLP | – | – | – |
| model.layers.8.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.8.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.8.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.8.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.8.self_attn | Qwen2Attention | – | – | – |
| model.layers.8.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.8.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.8.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.8.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.8.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.8.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.8.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.9 | Qwen2DecoderLayer | – | – | – |
| model.layers.9.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.9.mlp | Qwen2MLP | – | – | – |
| model.layers.9.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.9.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.9.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.9.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.9.self_attn | Qwen2Attention | – | – | – |
| model.layers.9.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.9.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.9.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.9.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.9.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.9.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.9.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.10 | Qwen2DecoderLayer | – | – | – |
| model.layers.10.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.10.mlp | Qwen2MLP | – | – | – |
| model.layers.10.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.10.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.10.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.10.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.10.self_attn | Qwen2Attention | – | – | – |
| model.layers.10.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.10.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.10.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.10.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.10.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.10.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.10.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.11 | Qwen2DecoderLayer | – | – | – |
| model.layers.11.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.11.mlp | Qwen2MLP | – | – | – |
| model.layers.11.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.11.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.11.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.11.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.11.self_attn | Qwen2Attention | – | – | – |
| model.layers.11.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.11.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.11.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.11.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.11.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.11.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |

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| Layer path | Type | Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.11.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.12 | Qwen2DecoderLayer | – | – | – |
| model.layers.12.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.12.mlp | Qwen2MLP | – | – | – |
| model.layers.12.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.12.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.12.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.12.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.12.self_attn | Qwen2Attention | – | – | – |
| model.layers.12.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.12.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.12.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.12.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.12.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.12.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.12.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.13 | Qwen2DecoderLayer | – | – | – |
| model.layers.13.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.13.mlp | Qwen2MLP | – | – | – |
| model.layers.13.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.13.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.13.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.13.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.13.self_attn | Qwen2Attention | – | – | – |
| model.layers.13.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.13.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.13.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.13.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.13.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.13.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.13.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.14 | Qwen2DecoderLayer | – | – | – |
| model.layers.14.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.14.mlp | Qwen2MLP | – | – | – |
| model.layers.14.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.14.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.14.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.14.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.14.self_attn | Qwen2Attention | – | – | – |
| model.layers.14.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.14.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.14.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.14.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.14.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.14.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.14.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.15 | Qwen2DecoderLayer | – | – | – |
| model.layers.15.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.15.mlp | Qwen2MLP | – | – | – |
| model.layers.15.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.15.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.15.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.15.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.15.self_attn | Qwen2Attention | – | – | – |
| model.layers.15.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.15.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.15.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.15.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.15.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.15.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.15.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.16 | Qwen2DecoderLayer | – | – | – |
| model.layers.16.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.16.mlp | Qwen2MLP | – | – | – |
| model.layers.16.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.16.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.16.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.16.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.16.self_attn | Qwen2Attention | – | – | – |

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| Layer path | Type | Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.16.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.16.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.16.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.16.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.16.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.16.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.16.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.17 | Qwen2DecoderLayer | – | – | – |
| model.layers.17.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.17.mlp | Qwen2MLP | – | – | – |
| model.layers.17.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.17.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.17.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.17.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.17.self_attn | Qwen2Attention | – | – | – |
| model.layers.17.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.17.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.17.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.17.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.17.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.17.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.17.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.18 | Qwen2DecoderLayer | – | – | – |
| model.layers.18.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.18.mlp | Qwen2MLP | – | – | – |
| model.layers.18.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.18.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.18.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.18.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.18.self_attn | Qwen2Attention | – | – | – |
| model.layers.18.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.18.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.18.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.18.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.18.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.18.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.18.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.19 | Qwen2DecoderLayer | – | – | – |
| model.layers.19.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.19.mlp | Qwen2MLP | – | – | – |
| model.layers.19.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.19.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.19.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.19.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.19.self_attn | Qwen2Attention | – | – | – |
| model.layers.19.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.19.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.19.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.19.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.19.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.19.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.19.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.20 | Qwen2DecoderLayer | – | – | – |
| model.layers.20.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.20.mlp | Qwen2MLP | – | – | – |
| model.layers.20.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.20.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.20.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.20.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.20.self_attn | Qwen2Attention | – | – | – |
| model.layers.20.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.20.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.20.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.20.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.20.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.20.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.20.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.21 | Qwen2DecoderLayer | – | – | – |
| model.layers.21.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.21.mlp | Qwen2MLP | – | – | – |

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| Layer path | Type | Shape | Dtype | Params |
|---|---|---|---|---|
| model.layers.21.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.21.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.21.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.21.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.21.self_attn | Qwen2Attention | – | – | – |
| model.layers.21.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.21.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.21.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.21.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.21.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.21.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.21.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.22 | Qwen2DecoderLayer | – | – | – |
| model.layers.22.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.22.mlp | Qwen2MLP | – | – | – |
| model.layers.22.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.22.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.22.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.22.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.22.self_attn | Qwen2Attention | – | – | – |
| model.layers.22.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.22.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.22.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.22.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.22.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.22.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.22.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.23 | Qwen2DecoderLayer | – | – | – |
| model.layers.23.input_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.23.mlp | Qwen2MLP | – | – | – |
| model.layers.23.mlp.down_proj.weight | Linear | 896 × 4864 | bfloat16 | 4,358,144 |
| model.layers.23.mlp.gate_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.23.mlp.up_proj.weight | Linear | 4864 × 896 | bfloat16 | 4,358,144 |
| model.layers.23.post_attention_layernorm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |
| model.layers.23.self_attn | Qwen2Attention | – | – | – |
| model.layers.23.self_attn.k_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.23.self_attn.k_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.layers.23.self_attn.o_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.23.self_attn.q_proj.bias | Linear | 896 | bfloat16 | 896 |
| model.layers.23.self_attn.q_proj.weight | Linear | 896 × 896 | bfloat16 | 802,816 |
| model.layers.23.self_attn.v_proj.bias | Linear | 128 | bfloat16 | 128 |
| model.layers.23.self_attn.v_proj.weight | Linear | 128 × 896 | bfloat16 | 114,688 |
| model.norm.weight | Qwen2RMSNorm | 896 | bfloat16 | 896 |

# Appendix B Per-Layer Parameter Count

| Module | Params | Running total |
|---|---|---|
| model.embed_tokens | 136,134,656 | 136,134,656 |
| model.layers.0.input_layernorm | 896 | 136,135,552 |
| model.layers.0.mlp.down_proj | 4,358,144 | 140,493,696 |
| model.layers.0.mlp.gate_proj | 4,358,144 | 144,851,840 |
| model.layers.0.mlp.up_proj | 4,358,144 | 149,209,984 |
| model.layers.0.post_attention_layernorm | 896 | 149,210,880 |
| model.layers.0.self_attn.k_proj | 114,816 | 149,325,696 |
| model.layers.0.self_attn.o_proj | 802,816 | 150,128,512 |
| model.layers.0.self_attn.q_proj | 803,712 | 150,932,224 |
| model.layers.0.self_attn.v_proj | 114,816 | 151,047,040 |
| model.layers.1.input_layernorm | 896 | 151,047,936 |
| model.layers.1.mlp.down_proj | 4,358,144 | 155,406,080 |
| model.layers.1.mlp.gate_proj | 4,358,144 | 159,764,224 |
| model.layers.1.mlp.up_proj | 4,358,144 | 164,122,368 |
| model.layers.1.post_attention_layernorm | 896 | 164,123,264 |
| model.layers.1.self_attn.k_proj | 114,816 | 164,238,080 |

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| Module | Params | Running total |
|---|---|---|
| model.layers.1.self_attn.o_proj | 802,816 | 165,040,896 |
| model.layers.1.self_attn.q_proj | 803,712 | 165,844,608 |
| model.layers.1.self_attn.v_proj | 114,816 | 165,959,424 |
| model.layers.2.input_layernorm | 896 | 165,960,320 |
| model.layers.2.mlp.down_proj | 4,358,144 | 170,318,464 |
| model.layers.2.mlp.gate_proj | 4,358,144 | 174,676,608 |
| model.layers.2.mlp.up_proj | 4,358,144 | 179,034,752 |
| model.layers.2.post_attention_layernorm | 896 | 179,035,648 |
| model.layers.2.self_attn.k_proj | 114,816 | 179,150,464 |
| model.layers.2.self_attn.o_proj | 802,816 | 179,953,280 |
| model.layers.2.self_attn.q_proj | 803,712 | 180,756,992 |
| model.layers.2.self_attn.v_proj | 114,816 | 180,871,808 |
| model.layers.3.input_layernorm | 896 | 180,872,704 |
| model.layers.3.mlp.down_proj | 4,358,144 | 185,230,848 |
| model.layers.3.mlp.gate_proj | 4,358,144 | 189,588,992 |
| model.layers.3.mlp.up_proj | 4,358,144 | 193,947,136 |
| model.layers.3.post_attention_layernorm | 896 | 193,948,032 |
| model.layers.3.self_attn.k_proj | 114,816 | 194,062,848 |
| model.layers.3.self_attn.o_proj | 802,816 | 194,865,664 |
| model.layers.3.self_attn.q_proj | 803,712 | 195,669,376 |
| model.layers.3.self_attn.v_proj | 114,816 | 195,784,192 |
| model.layers.4.input_layernorm | 896 | 195,785,088 |
| model.layers.4.mlp.down_proj | 4,358,144 | 200,143,232 |
| model.layers.4.mlp.gate_proj | 4,358,144 | 204,501,376 |
| model.layers.4.mlp.up_proj | 4,358,144 | 208,859,520 |
| model.layers.4.post_attention_layernorm | 896 | 208,860,416 |
| model.layers.4.self_attn.k_proj | 114,816 | 208,975,232 |
| model.layers.4.self_attn.o_proj | 802,816 | 209,778,048 |
| model.layers.4.self_attn.q_proj | 803,712 | 210,581,760 |
| model.layers.4.self_attn.v_proj | 114,816 | 210,696,576 |
| model.layers.5.input_layernorm | 896 | 210,697,472 |
| model.layers.5.mlp.down_proj | 4,358,144 | 215,055,616 |
| model.layers.5.mlp.gate_proj | 4,358,144 | 219,413,760 |
| model.layers.5.mlp.up_proj | 4,358,144 | 223,771,904 |
| model.layers.5.post_attention_layernorm | 896 | 223,772,800 |
| model.layers.5.self_attn.k_proj | 114,816 | 223,887,616 |
| model.layers.5.self_attn.o_proj | 802,816 | 224,690,432 |
| model.layers.5.self_attn.q_proj | 803,712 | 225,494,144 |
| model.layers.5.self_attn.v_proj | 114,816 | 225,608,960 |
| model.layers.6.input_layernorm | 896 | 225,609,856 |
| model.layers.6.mlp.down_proj | 4,358,144 | 229,968,000 |
| model.layers.6.mlp.gate_proj | 4,358,144 | 234,326,144 |
| model.layers.6.mlp.up_proj | 4,358,144 | 238,684,288 |
| model.layers.6.post_attention_layernorm | 896 | 238,685,184 |
| model.layers.6.self_attn.k_proj | 114,816 | 238,800,000 |
| model.layers.6.self_attn.o_proj | 802,816 | 239,602,816 |
| model.layers.6.self_attn.q_proj | 803,712 | 240,406,528 |
| model.layers.6.self_attn.v_proj | 114,816 | 240,521,344 |
| model.layers.7.input_layernorm | 896 | 240,522,240 |
| model.layers.7.mlp.down_proj | 4,358,144 | 244,880,384 |
| model.layers.7.mlp.gate_proj | 4,358,144 | 249,238,528 |
| model.layers.7.mlp.up_proj | 4,358,144 | 253,596,672 |
| model.layers.7.post_attention_layernorm | 896 | 253,597,568 |
| model.layers.7.self_attn.k_proj | 114,816 | 253,712,384 |
| model.layers.7.self_attn.o_proj | 802,816 | 254,515,200 |
| model.layers.7.self_attn.q_proj | 803,712 | 255,318,912 |
| model.layers.7.self_attn.v_proj | 114,816 | 255,433,728 |
| model.layers.8.input_layernorm | 896 | 255,434,624 |

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| Module | Params | Running total |
|---|---|---|
| model.layers.8.mlp.down_proj | 4,358,144 | 259,792,768 |
| model.layers.8.mlp.gate_proj | 4,358,144 | 264,150,912 |
| model.layers.8.mlp.up_proj | 4,358,144 | 268,509,056 |
| model.layers.8.post_attention_layernorm | 896 | 268,509,952 |
| model.layers.8.self_attn.k_proj | 114,816 | 268,624,768 |
| model.layers.8.self_attn.o_proj | 802,816 | 269,427,584 |
| model.layers.8.self_attn.q_proj | 803,712 | 270,231,296 |
| model.layers.8.self_attn.v_proj | 114,816 | 270,346,112 |
| model.layers.9.input_layernorm | 896 | 270,347,008 |
| model.layers.9.mlp.down_proj | 4,358,144 | 274,705,152 |
| model.layers.9.mlp.gate_proj | 4,358,144 | 279,063,296 |
| model.layers.9.mlp.up_proj | 4,358,144 | 283,421,440 |
| model.layers.9.post_attention_layernorm | 896 | 283,422,336 |
| model.layers.9.self_attn.k_proj | 114,816 | 283,537,152 |
| model.layers.9.self_attn.o_proj | 802,816 | 284,339,968 |
| model.layers.9.self_attn.q_proj | 803,712 | 285,143,680 |
| model.layers.9.self_attn.v_proj | 114,816 | 285,258,496 |
| model.layers.10.input_layernorm | 896 | 285,259,392 |
| model.layers.10.mlp.down_proj | 4,358,144 | 289,617,536 |
| model.layers.10.mlp.gate_proj | 4,358,144 | 293,975,680 |
| model.layers.10.mlp.up_proj | 4,358,144 | 298,333,824 |
| model.layers.10.post_attention_layernorm | 896 | 298,334,720 |
| model.layers.10.self_attn.k_proj | 114,816 | 298,449,536 |
| model.layers.10.self_attn.o_proj | 802,816 | 299,252,352 |
| model.layers.10.self_attn.q_proj | 803,712 | 300,056,064 |
| model.layers.10.self_attn.v_proj | 114,816 | 300,170,880 |
| model.layers.11.input_layernorm | 896 | 300,171,776 |
| model.layers.11.mlp.down_proj | 4,358,144 | 304,529,920 |
| model.layers.11.mlp.gate_proj | 4,358,144 | 308,888,064 |
| model.layers.11.mlp.up_proj | 4,358,144 | 313,246,208 |
| model.layers.11.post_attention_layernorm | 896 | 313,247,104 |
| model.layers.11.self_attn.k_proj | 114,816 | 313,361,920 |
| model.layers.11.self_attn.o_proj | 802,816 | 314,164,736 |
| model.layers.11.self_attn.q_proj | 803,712 | 314,968,448 |
| model.layers.11.self_attn.v_proj | 114,816 | 315,083,264 |
| model.layers.12.input_layernorm | 896 | 315,084,160 |
| model.layers.12.mlp.down_proj | 4,358,144 | 319,442,304 |
| model.layers.12.mlp.gate_proj | 4,358,144 | 323,800,448 |
| model.layers.12.mlp.up_proj | 4,358,144 | 328,158,592 |
| model.layers.12.post_attention_layernorm | 896 | 328,159,488 |
| model.layers.12.self_attn.k_proj | 114,816 | 328,274,304 |
| model.layers.12.self_attn.o_proj | 802,816 | 329,077,120 |
| model.layers.12.self_attn.q_proj | 803,712 | 329,880,832 |
| model.layers.12.self_attn.v_proj | 114,816 | 329,995,648 |
| model.layers.13.input_layernorm | 896 | 329,996,544 |
| model.layers.13.mlp.down_proj | 4,358,144 | 334,354,688 |
| model.layers.13.mlp.gate_proj | 4,358,144 | 338,712,832 |
| model.layers.13.mlp.up_proj | 4,358,144 | 343,070,976 |
| model.layers.13.post_attention_layernorm | 896 | 343,071,872 |
| model.layers.13.self_attn.k_proj | 114,816 | 343,186,688 |
| model.layers.13.self_attn.o_proj | 802,816 | 343,989,504 |
| model.layers.13.self_attn.q_proj | 803,712 | 344,793,216 |
| model.layers.13.self_attn.v_proj | 114,816 | 344,908,032 |
| model.layers.14.input_layernorm | 896 | 344,908,928 |
| model.layers.14.mlp.down_proj | 4,358,144 | 349,267,072 |
| model.layers.14.mlp.gate_proj | 4,358,144 | 353,625,216 |
| model.layers.14.mlp.up_proj | 4,358,144 | 357,983,360 |
| model.layers.14.post_attention_layernorm | 896 | 357,984,256 |

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| Module | Params | Running total |
|---|---|---|
| model.layers.14.self_attn.k_proj | 114,816 | 358,099,072 |
| model.layers.14.self_attn.o_proj | 802,816 | 358,901,888 |
| model.layers.14.self_attn.q_proj | 803,712 | 359,705,600 |
| model.layers.14.self_attn.v_proj | 114,816 | 359,820,416 |
| model.layers.15.input_layernorm | 896 | 359,821,312 |
| model.layers.15.mlp.down_proj | 4,358,144 | 364,179,456 |
| model.layers.15.mlp.gate_proj | 4,358,144 | 368,537,600 |
| model.layers.15.mlp.up_proj | 4,358,144 | 372,895,744 |
| model.layers.15.post_attention_layernorm | 896 | 372,896,640 |
| model.layers.15.self_attn.k_proj | 114,816 | 373,011,456 |
| model.layers.15.self_attn.o_proj | 802,816 | 373,814,272 |
| model.layers.15.self_attn.q_proj | 803,712 | 374,617,984 |
| model.layers.15.self_attn.v_proj | 114,816 | 374,732,800 |
| model.layers.16.input_layernorm | 896 | 374,733,696 |
| model.layers.16.mlp.down_proj | 4,358,144 | 379,091,840 |
| model.layers.16.mlp.gate_proj | 4,358,144 | 383,449,984 |
| model.layers.16.mlp.up_proj | 4,358,144 | 387,808,128 |
| model.layers.16.post_attention_layernorm | 896 | 387,809,024 |
| model.layers.16.self_attn.k_proj | 114,816 | 387,923,840 |
| model.layers.16.self_attn.o_proj | 802,816 | 388,726,656 |
| model.layers.16.self_attn.q_proj | 803,712 | 389,530,368 |
| model.layers.16.self_attn.v_proj | 114,816 | 389,645,184 |
| model.layers.17.input_layernorm | 896 | 389,646,080 |
| model.layers.17.mlp.down_proj | 4,358,144 | 394,004,224 |
| model.layers.17.mlp.gate_proj | 4,358,144 | 398,362,368 |
| model.layers.17.mlp.up_proj | 4,358,144 | 402,720,512 |
| model.layers.17.post_attention_layernorm | 896 | 402,721,408 |
| model.layers.17.self_attn.k_proj | 114,816 | 402,836,224 |
| model.layers.17.self_attn.o_proj | 802,816 | 403,639,040 |
| model.layers.17.self_attn.q_proj | 803,712 | 404,442,752 |
| model.layers.17.self_attn.v_proj | 114,816 | 404,557,568 |
| model.layers.18.input_layernorm | 896 | 404,558,464 |
| model.layers.18.mlp.down_proj | 4,358,144 | 408,916,608 |
| model.layers.18.mlp.gate_proj | 4,358,144 | 413,274,752 |
| model.layers.18.mlp.up_proj | 4,358,144 | 417,632,896 |
| model.layers.18.post_attention_layernorm | 896 | 417,633,792 |
| model.layers.18.self_attn.k_proj | 114,816 | 417,748,608 |
| model.layers.18.self_attn.o_proj | 802,816 | 418,551,424 |
| model.layers.18.self_attn.q_proj | 803,712 | 419,355,136 |
| model.layers.18.self_attn.v_proj | 114,816 | 419,469,952 |
| model.layers.19.input_layernorm | 896 | 419,470,848 |
| model.layers.19.mlp.down_proj | 4,358,144 | 423,828,992 |
| model.layers.19.mlp.gate_proj | 4,358,144 | 428,187,136 |
| model.layers.19.mlp.up_proj | 4,358,144 | 432,545,280 |
| model.layers.19.post_attention_layernorm | 896 | 432,546,176 |
| model.layers.19.self_attn.k_proj | 114,816 | 432,660,992 |
| model.layers.19.self_attn.o_proj | 802,816 | 433,463,808 |
| model.layers.19.self_attn.q_proj | 803,712 | 434,267,520 |
| model.layers.19.self_attn.v_proj | 114,816 | 434,382,336 |
| model.layers.20.input_layernorm | 896 | 434,383,232 |
| model.layers.20.mlp.down_proj | 4,358,144 | 438,741,376 |
| model.layers.20.mlp.gate_proj | 4,358,144 | 443,099,520 |
| model.layers.20.mlp.up_proj | 4,358,144 | 447,457,664 |
| model.layers.20.post_attention_layernorm | 896 | 447,458,560 |
| model.layers.20.self_attn.k_proj | 114,816 | 447,573,376 |
| model.layers.20.self_attn.o_proj | 802,816 | 448,376,192 |
| model.layers.20.self_attn.q_proj | 803,712 | 449,179,904 |
| model.layers.20.self_attn.v_proj | 114,816 | 449,294,720 |

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| Module | Params | Running total |
|---|---|---|
| model.layers.21.input_layernorm | 896 | 449,295,616 |
| model.layers.21.mlp.down_proj | 4,358,144 | 453,653,760 |
| model.layers.21.mlp.gate_proj | 4,358,144 | 458,011,904 |
| model.layers.21.mlp.up_proj | 4,358,144 | 462,370,048 |
| model.layers.21.post_attention_layernorm | 896 | 462,370,944 |
| model.layers.21.self_attn.k_proj | 114,816 | 462,485,760 |
| model.layers.21.self_attn.o_proj | 802,816 | 463,288,576 |
| model.layers.21.self_attn.q_proj | 803,712 | 464,092,288 |
| model.layers.21.self_attn.v_proj | 114,816 | 464,207,104 |
| model.layers.22.input_layernorm | 896 | 464,208,000 |
| model.layers.22.mlp.down_proj | 4,358,144 | 468,566,144 |
| model.layers.22.mlp.gate_proj | 4,358,144 | 472,924,288 |
| model.layers.22.mlp.up_proj | 4,358,144 | 477,282,432 |
| model.layers.22.post_attention_layernorm | 896 | 477,283,328 |
| model.layers.22.self_attn.k_proj | 114,816 | 477,398,144 |
| model.layers.22.self_attn.o_proj | 802,816 | 478,200,960 |
| model.layers.22.self_attn.q_proj | 803,712 | 479,004,672 |
| model.layers.22.self_attn.v_proj | 114,816 | 479,119,488 |
| model.layers.23.input_layernorm | 896 | 479,120,384 |
| model.layers.23.mlp.down_proj | 4,358,144 | 483,478,528 |
| model.layers.23.mlp.gate_proj | 4,358,144 | 487,836,672 |
| model.layers.23.mlp.up_proj | 4,358,144 | 492,194,816 |
| model.layers.23.post_attention_layernorm | 896 | 492,195,712 |
| model.layers.23.self_attn.k_proj | 114,816 | 492,310,528 |
| model.layers.23.self_attn.o_proj | 802,816 | 493,113,344 |
| model.layers.23.self_attn.q_proj | 803,712 | 493,917,056 |
| model.layers.23.self_attn.v_proj | 114,816 | 494,031,872 |
| model.norm | 896 | 494,032,768 |